

Understanding Java Code Execution

Step-by-Step Walkthrough

Programming Tutorial

Code Execution Flow

Line 1: Define the Main class

```
2 public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello World: 1");  
    }  
}
```

Code Execution Flow

Line 2: Define the main method entry point

```
public class Main {
```

3

```
    public static void main(String[] args) {
```

```
        System.out.println("Hello World: 1");
```

```
        System.out.println("Hello World: 2");
```

Code Execution Flow

Line 3: Print the first hello message

```
public class Main {  
    public static void main(String[] args) {  
4      System.out.println("Hello World: 1");  
        System.out.println("Hello World: 2");  
        System.out.println("Hello World: 3");  
    }  
}
```

Code Execution Flow

Line 4: Print the second hello message

```
public static void main(String[] args) {  
    System.out.println("Hello World: 1");  
    System.out.println("Hello World: 2");  
    System.out.println("Hello World: 3");  
    System.out.println("Hello World: 4");  
}
```

5

Code Execution Flow

Line 5: Print the third hello message

```
System.out.println("Hello World: 1");  
System.out.println("Hello World: 2");  
6 System.out.println("Hello World: 3");  
System.out.println("Hello World: 4");  
System.out.println("Hello World: 5");
```

Code Execution Flow

Line 6: Print the fourth hello message

```
System.out.println("Hello World: 2");  
System.out.println("Hello World: 3");  
7 System.out.println("Hello World: 4");  
System.out.println("Hello World: 5");  
System.out.println("Hello World: 6");
```

Code Execution Flow

Line 7: Close the main method

```
System.out.println("Hello World: 3");  
System.out.println("Hello World: 4");  
8 System.out.println("Hello World: 5");  
System.out.println("Hello World: 6");  
System.out.println("Hello World: 7");
```

Code Execution Flow

Line 8: Close the Main class

```
System.out.println("Hello World: 4");  
System.out.println("Hello World: 5");  
9 System.out.println("Hello World: 6");  
System.out.println("Hello World: 7");  
System.out.println("Hello World: 8");
```

Code Execution Flow

Line 9: Executing line 9

```
10 System.out.println("Hello World: 5");  
    System.out.println("Hello World: 6");  
    System.out.println("Hello World: 7");  
    System.out.println("Hello World: 8");  
    }
```

Code Execution Flow

Line 10: Executing line 10

```
System.out.println("Hello World: 6");  
System.out.println("Hello World: 7");  
11 System.out.println("Hello World: 8");  
    }
```

Code Execution Flow

Line 11: Executing line 11

```
System.out.println("Hello World: 7");  
System.out.println("Hello World: 8");
```

```
12      }
```